

(TR-104) PYTHON WITH AI/ML

Training Day 18 Report

Date: 4 February 2026

Topic Covered: Multithreading, Multiprocessing & Performance Optimization in Python

The eighteenth day of training focused on understanding multithreading vs multiprocessing concepts in Python along with performance optimization techniques. The day included hands-on implementation of concurrent execution and performance profiling.

- Studied the difference between multithreading and multiprocessing.
- Understood the concept of GIL (Global Interpreter Lock).
- Learned CPU-bound vs I/O-bound tasks and their execution behavior.
- Worked with Python threading module for concurrent execution.
- Used multiprocessing module for parallel processing tasks.
- Developed a threaded file downloader using multiple threads.
- Implemented a multiprocessing-based image resizing script.
- Performed performance profiling using cProfile module.
- Analyzed execution time using time and timeit modules.
- Practiced optimization of loops and data structures for better performance.
- Explored memory optimization concepts including __slots__.

- **GitHub Link:**

https://github.com/JaspinderKaurWalia26/Python_Practice_Codes/tree/main/04-02-2026

- **Blog Link:**

<https://medium.com/@jaspinderkaurwalia855/multithreading-vs-multiprocessing-in-python-bcb37db1d5fe>